1. Elevator Pitch

Our Panoramic Treadmill + AI Metabolism Precision Intervention System addresses the global teen obesity crisis, targeting poor self-control, boring home workouts, and disconnected health data. Integrating panoramic simulation, AI metabolic monitoring, visualized real-time data, a knee-protective water-surrounded design (using buoyancy and resistance for safety and muscle engagement), and interactive competition, it delivers personalized, engaging solutions. Unlike traditional treadmills, it links exercise, physical, and dietary data in a closed loop with safety monitoring. Targeting buyers aged 25-50 (parents, fitness enthusiasts) and semi-professional runners as initial users, it meets urgent teen obesity intervention needs and professional home fitness demands, unlocking commercial potential through product sales and data-driven services.

2. Team

Composed of three high school students and one advisor, our team is driven by research on Prader-Willi Syndrome—we aim to support this special group and other obese populations through innovation. Divisions are clear: one student leads product design and programming (mechanical structure, projection debugging, APP core coding); two focus on marketing (market research, demand analysis, social media promotion); the advisor, with engineering and project experience, reviews flaws, supplements technical details, and optimizes logic. Leveraging a student's insight into teen pain points and the advisor's professional guidance, we ensure the product is scientifically sound and user-centric, empowering both special and general obese adolescents.

3. Opportunity

Global teen obesity is a public health emergency: 5-19-year-old obesity rates surged 8.3 times (1975-2022), with 100+ countries exceeding 15%; China's 6-17-year-old rate hit 12.1% in 2022. Consequences include 37.2% of obese teens having fatty liver, a 24-fold rise in 12-14-year-old type 2 diabetes, and 11.3-point higher depression scores. Current solutions fail: traditional treadmills lack data linkage and simulation; home workouts are boring (low adherence); weight management relies on inconsistent family supervision. Parents (25-50) need safe, engaging tools, while runners seek data-driven training equipment. The intersection of the obesity crisis, unmet family needs, and professional fitness demand creates a massive opportunity. Our product fills the gap with precision monitoring, immersion, and safety, offering a teen-friendly obesity intervention solution.

4. Key Metrics

Innovation, Design, and Technology

Our system features five core functions:

1. Panoramic Simulation: 5G + 3D holography recreates beaches, snowfields, and plateaus. A closed, deformable cabin adjusts slope, wind speed, temperature, and oxygen—e.g., low-temperature snowfields for sugar metabolism, low-oxygen plateaus for fatty acid oxidation—delivering immersive experiences.
2. Metabolism Monitoring: Sensors track height, weight, heart rate, blood oxygen, and respiration. AI analyzes sugar/lipid metabolism, combines with diet records to generate personalized plans, and provides visualized feedback on progress.
3. Water-Surrounded Design: A water base reduces knee impact by 40% (vs. traditional

treadmills) via buoyancy and boosts muscle engagement with water resistance, balancing safety and effectiveness.

4. Interactive Competition: 5G enables real-time online races with friends' holographic projections, fostering communication and adherence.
5. Safety Monitoring: Ultrasonic and touch sensors detect falls/emergencies, triggering automatic cabin opening and APP alerts to guardians.

Novelty and Proprietary Attributes

Virtual-Real Metabolism Optimization: Tailors environments/plans to individual metabolic types, boosting weight loss efficiency by 30%.

Closed-Loop Data: Integrates exercise, physical, diet, and metabolic data, solving fragmentation with end-to-end feedback.

Teen-Centric Design: Interactive competition and immersion raise adherence from 30% (traditional treadmills) to 70%.

Need Fulfillment

Teens: Reduced boredom, lower injury risk, and clear progress feedback.

Parents: Real-time monitoring, safety alerts, and AI plans eliminate constant supervision.

Runners: Professional data and environment simulation support targeted training.

Impact

Individual: 15-20% weight loss in 3 months for obese teens, 40% lower knee injury risk, 25% improved metabolic health; runners' 5km bests up 5-8%.

Social: Reduces long-term healthcare burdens (21% of U.S. health spending is obesity-related) and lowers adult chronic disease rates.

Protection

Patents for water-surrounded structure, metabolic algorithm, and simulation system.

Trade secrets for AI codes and models; copyrights for software/APP design; brand loyalty and partnerships build barriers.

5. Validation/Progress

Validation

1. Market Research: WHO, *The Lancet*, and China CDC data confirm the crisis. A survey of 500 parents (25-50) and 30 semi-professional runners shows 83% of parents will pay $800-$1,500 for data-driven teen fitness equipment, and 78% of runners want simulation/monitoring features.
2. Technical Feasibility: Metabolic algorithm tests with 30 obese teens show 92% accuracy; water design reduces knee impact by 40%; 5G projection achieves lag-free interaction.
3. User Feedback: Beta tests with 20 teens and 10 runners: 65% higher teen satisfaction, weekly exercise up from 1.2 to 3.5 times; 90% runner satisfaction with data/simulation.

Progress

1. Product Development: Created operational and ideal models; designed data synchronization

and basic APP structure.

2. Intellectual Property: Filed 3 utility model patents and 2 software copyrights (under review).

Reference Attachment

1. WHO Global Observatory for Youth Health. (2023). Global Youth Obesity Data.
2. *The Lancet* Child & Adolescent Health. (2022). Adolescent Obesity & Chronic Disease Trends.
3. Chinese CDC. (2022). *Report on Nutrition and Chronic Diseases of Chinese Residents*.
4. 2024 Survey Data (500 Parents, 30 Semi-Professional Runners).
5. 2024 Biomechanical Test Report (Water-Surrounded Design).

6. Market

Customers and Segments

Core Buyers: Middle-to-high-income parents (25-50) concerned about children's obesity.

Initial Pilot Users: Semi-professional runners (18-35)—positive feedback drives professional runner/parent adoption.

Secondary Users: Professional runners, fitness enthusiasts, schools, and community health centers.

Key Needs

Parents prioritize safety, effectiveness, fun, and data transparency; runners value data accuracy, simulation, and performance improvement.

Market Size

Global home fitness equipment market to reach $28.7B by 2027 (CAGR 6.8%); teen obesity intervention grows 12% annually. China has 30M obese 6-17-year-olds; professional home treadmill market is $5.2B, with unmet demand for data-driven immersive products.

Buyers vs. Payers

Aligned (parents buy for teens; runners self-purchase) with clear motivation and short decision-making.

Ecosystem

Includes manufacturers, 5G/AI providers, sports medicine institutions, social media (marketing), and logistics partners. Key stakeholders: end users, manufacturers, suppliers, and channels.

7. Competition

Competitors

1. Traditional Brands (e.g., NordicTrack): Basic speed/incline adjustment, limited data—no simulation/data linkage.
2. Smart Screen Brands (e.g., Peloton): Multimedia/online classes, no metabolic monitoring—high price ($1,500+).
3. Water Treadmills: Low-impact, no simulation/AI/interaction—for rehabilitation, not teens.
4. Fitness Apps (e.g., Keep): Exercise plans, no integrated monitoring/immersion—relies on external equipment.

Advantages and Disadvantages

Advantages

1. Virtual-real metabolic optimization (individual adaptation)
2. Panoramic simulation + interactive competition (high teen adherence)
3. Water design (knee protection + muscle training)
4. Closed-loop data + AI plans (quantifiable results)
5. Safety monitoring + parental alerts

Disadvantages

1. Higher R&D/production costs than traditional treadmills
2. 5G-dependent (limited remote area experience)
3. New brand (lower awareness than incumbents)

Positioning

"Precision Health Partner for Teens + Professional Training Tool for Runners"—combining medical-grade monitoring, consumer-friendly experience, and professional performance. Differentiated by obesity intervention, safety, and immersion, targeting middle-to-high-income families and fitness enthusiasts.

8. Go-To-Market

We use an integrated multi-channel strategy:

1. Influencer Marketing: Collaborate with semi-professional runners and parent KOLs on TikTok for reviews/results to drive initial conversion.
2. Pilot Programs: Recruit 50 semi-professional runners for 3-month beta tests—collect feedback for word-of-mouth to reach professionals/parents.
3. Omnichannel Sales: Sell via official websites, Amazon, JD.com, and offline stores—offer courier delivery/in-store pickup plus personalized installation.
4. Strategic Partnerships: Collaborate with schools/community health centers on teen obesity programs to expand B2B channels and brand influence.

9. Business Model

Revenue Streams

1. Product Sales: Treadmills priced $1,200-$1,800 (mid-range, feature-rich). Target: 10,000 units (Year 1), 30,000 units (Year 2).
2. Subscription Services: $10-$15/month premium membership (AI plan updates, personalized diet, professional courses)—30% buyer conversion expected.
3. B2B Cooperation: Bulk purchase licensing for schools/health centers (custom health dashboards)—license fees.

Cost Structure

1. Production: $600-$800/unit (materials, production: 30%, logistics: 10%).
2. R&D: $500,000/year (algorithm optimization, software updates).
3. Marketing: $300,000/year (ads, influencer collaborations, pilots).

4. Operations: $200,000/year (salaries, customer service, APP maintenance).

Unit Economics

Unit Price: $1,500 (average)

Unit Cost: $700 (production + logistics)

Gross Margin: ~53%

Subscription Margin: 85% (low variable costs)

10. Fundraising

We seek $1.5M seed funding:

$600k: Product optimization/mass production (mold development, component procurement, quality control).

$400k: Marketing (social media, influencer partnerships, pilots) for initial sales/brand exposure.

$300k: R&D (algorithm/APP updates, new features) to maintain competitiveness.

$200k: Operational reserves (staffing, customer service, logistics).

Total budget: $1.5M. Funding sources: fitness/health tech angels, consumer tech VCs, and public health innovation grants—aligned with project mission.